

Aggregate functions

- Consolidates our data
- Summarize our data
- Combines
- In SQL aggregate functions summarize many rows into one result (or one result per group)

(1) COUNT() → how many rows

(2) SUM() → total of a numeric column

(3) AVG() → average of a numeric column

(4) MIN() → smallest value

(5) MAX() → largest value

- An aggregate looks at multiple rows in a table & returns a single value

Products		
ID	fruitname	Revenue
101	Apple	2
102	Banana	4
103	orange	3
104	Kiwi	5

● Select SUM(Revenue) AS total-revenue
from Products;

Select MAX(Revenue) AS highest-revenue
from products;

highest revenue
5

● Select MIN(Revenue) AS lowest-revenue
from products;

lowest revenue
2

Looks for the smallest or minimum value
in the revenue column

Select COUNT(ID) AS number-of-fruits
from products;

Number-of-fruits
4

An aggregate function

Male	M	Grade 10H
F	M	
M	F	
F	M	

Group BY →

ID	Name	Gender
101	Mauli	F
102	Rofi	M
103	Phobo	m
104	Jacob	m
105	John	m
106	Leralo	F
107	Robert	m
108	Anna	F

Write a Spl script to count number of
males & number of females

Select COUNT(ID) AS number-of-students,
gender
from Grade 10M
Group By Gender;

Number-of-students	Gender
5	M
3	F

What is an aggregate function

- It summarizes data into one result
- Is a function that performs a calculation on multiple rows of data & returns a single result
- Perform calculation in our rows & returns one result or one result per group.
- Aggregate functions are commonly used with the GROUP BY statement to group rows that belong to the same category

Syntax of GROUP BY

SELECT Aggregate function (column1) AS descriptive-name
categorical-column
FROM table-name
GROUP BY categorical-column;

- We only use the GROUP BY statement when we are aggregating the data & in our SELECT we have a categorical column

- we do not GROUP BY the aggregate column

what is a categorical column?

- Is a column that gives us a classification of specific data eg Gender

COUNT() - we commonly count the unique identifiers (ID)

SELECT Count (ID) AS number-of-students,
Gender
FROM Cradelott
GROUP BY gender;

Number-of-Students	Gender
5	M
3	F

Sales_id	Store	Amount	You cant group by without having a categorical column in the select
1	Mr Price	100	
2	Mr Price	50	
3	Pep	80	
4	Pep	20	

- 1- Write a SQL query to calculate the total revenue generated by all stores
- 2- Write a SQL query to the total revenue per store group

Select count(Amount) AS total-revenue,
 from Sales
 Group by store;

Total-revenue	store Amount
150	Mr Price
100	Pep

• Select SUM(amount) AS total-revenue,
 from Sales;

Total-revenue
250

store,
 Select, SUM(amount) AS total-revenue
 from Sales
 Group BY store;

→ Do not perform SUM() on an ID or
 a unique identifier.

store	Total-revenue
Mr Price	150
Pep	100